

Fewer than 3.178^n stable matchings

Proof and formal verification note

25 September 2026

Abstract

For every positive integer n , a strict, complete stable-marriage instance with n men and n women has fewer than $(1589/500)^n = 3.178^n$ stable matchings. The proof bounds the entropy of the stable matching family through a random revelation order. Propagation in both directions of each stable-partner list reduces the conditional support; a simultaneous-cap remainder supplies an additional saving. A finite charging argument accounts for exceptional vertices, and rational logarithm certificates establish the final constant. This note explains the argument accompanying the complete Lean proof. The theorem has no certificate hypothesis or restriction on preference patterns.

1 The theorem

A profile is *strict and complete* if every participant strictly ranks every participant on the other side. A matching is a bijection between the two sides. A pair blocks a matching when both participants strictly prefer each other to their assigned partners; a matching is *stable* if it has no blocking pair. Let $\mathcal{S}(I)$ be the family of all stable matchings of a profile I , and put

$$f(n) = \max_I |\mathcal{S}(I)|,$$

where the maximum ranges over strict, complete profiles with n participants on each side.

Theorem 1. *For every positive integer n and every strict, complete profile I with n men and n women,*

$$|\mathcal{S}(I)| < \left(\frac{1589}{500}\right)^n = 3.178^n. \tag{1.1}$$

In particular, $f(n) < 3.178^n$ for all $n > 0$.

The rational base is exact. The result applies to the full stable matching family, with no bound on n and no structural hypothesis on rotations or preferences. The condition $n > 0$ is needed for the strict inequality: at $n = 0$ there is one empty matching.

The Lean theorem is `SmpMax.General.stableCount_lt_3178_pow`. Its definitions encode preference rows as permutations of ranks, with lower ranks preferred, and count every stable bijection. Appendix A gives the exact declaration and verification commands. The complete machine proof is the accompanying source; this note makes no optimality or priority claim.

2 Entropy and a common random gap

Assume $\mathcal{S}(I) \neq \emptyset$, since otherwise the theorem is immediate. Choose a reference matching μ uniformly from $\mathcal{S}(I)$. For a reveal order π , expose the men's partners one coordinate at a time. Let $X_m(\mu, \pi)$ be the number of partners of m occurring among stable matchings that agree with μ at every already exposed coordinate. The reference itself is compatible, so $X_m \geq 1$.

Lemma 1 (Finite revelation inequality). *For any independently chosen finite random reveal order,*

$$\log |\mathcal{S}(I)| \leq \mathbb{E}_{\mu, \pi} \left[\sum_m \log X_m(\mu, \pi) \right]. \quad (2.1)$$

Proof. For a fixed order, apply the chain rule for entropy to the uniformly chosen matching. Each conditional entropy is at most the logarithm of its conditional support size. Sum over the n coordinates and then average over the independently chosen order. $\square$

The formal development establishes the corresponding finite counting inequality, including the empty-family case. Auxiliary labels introduced below affect the reveal ranks only; the sum in (2.1) always has exactly n terms.

2.1 Padded stable-partner lists

Fix μ and m . Order all stable partners of m from best to worst. Label a partner w by her reference owner $\mu^{-1}(w)$. The labels are distinct, and the reference partner $\mu(m)$ has label m , the pivot. Stable-pair opposition implies that a previously revealed owner on either side bars compatible partners farther beyond that owner.

Append 24 auxiliary labels at each end of the list. Choose π uniformly from the permutations of the n real labels together with these 48 auxiliary labels. Every man uses the same auxiliary labels and the same permutation. Each individual padded list has distinct labels; independence between men's costs is unnecessary.

Let L and R count consecutive later-revealed positions on the two sides of the pivot, stopping at an earlier label or the padded end. The common gap has size

$$K = 1 + L + R. \quad (2.2)$$

Every compatible stable partner lies in this gap. Finite permutation counts and a telescoping logarithmic sum give the following uniform bound, valid independently of the list length.

Lemma 2 (Finite gap bound). *For a positive cutoff N , define*

$$B_N = \sum_{k=2}^N \frac{2}{k+1} \log \frac{k}{k-1} + \frac{1}{2N} + \frac{3}{2(N+1)}. \quad (2.3)$$

Then $\mathbb{E}_\pi[\log K] \leq B_N$. At cutoff $N = 512$,

$$B_{512} < \log \frac{1665987}{500000} = \log 3.331974. \quad (2.4)$$

The tail estimate is finite, and (2.4) is proved by an exact rational certificate. The argument needs no identification of B_N with an infinite-series limit.

3 Propagation and directional savings

Fix one direction of each stable-partner list. Let $g(m)$ be the reference owner of the nearest partner in that direction, or m if no such partner exists. Crossing a worse stable partner forces her reference owner to become worse off, and that change propagates to the owner's nearest worse-partner owner. The corresponding implication toward better partners follows from stable-pair opposition in the original profile.

Thus crossing a partner owned by a forces $g(a)$ to change. If $g(a)$ has already been revealed, compatibility with μ is contradicted. The crossed partner and every farther partner are excluded. Both directions are proved for the original preferences.

Let l, r be the actual left and right compatible-support caps about the reference partner. They satisfy

$$0 \leq l \leq L, \quad 0 \leq r \leq R, \quad 1 \leq X_m \leq T := 1 + l + r. \quad (3.1)$$

The compatible support need not fill the interval between its extremes; its cardinality is bounded by the interval length. Measure the two directional losses against the same gap:

$$D_L = \log K - \log(1 + l + R), \quad D_R = \log K - \log(1 + L + r). \quad (3.2)$$

Both losses are nonnegative. Write

$$a_j = \log \frac{j+2}{j+1}, \quad \delta = \frac{467}{20000}, \quad \varepsilon = \frac{307}{500000}. \quad (3.3)$$

3.1 Short sides and noncycles

If a side has at most one real stable partner, its cap is at most one. Five padded-gap contributions, telescoped before averaging where they are nested, yield a directional saving of at least

$$\alpha = \frac{a_1}{12} + \frac{a_2}{10} + \frac{a_3}{15} \geq 3(\delta + \varepsilon). \quad (3.4)$$

If $g^2(m) \neq m$, the first partner has a propagation witness outside the pivot and first neighbor. Three disjoint events, with opposite gap lengths 0, 1, and 2, give at least

$$\beta = \frac{a_0}{12} + \frac{a_1}{30} + \frac{a_2}{60} \geq 3(\delta + \varepsilon). \quad (3.5)$$

The possible coincidences of that witness with the first three opposite labels are treated separately; each gives the same certified lower bound or a larger one.

3.2 Regular second-step events

Suppose the chosen side has at least two stable partners and $g^2(m) = m$. Let b own the second partner. In the regular case, $u = g(b)$ differs from the pivot, first neighbor, and second neighbor. If u is revealed before the pivot, the chosen-side cap is at most one.

For $j = 0, \dots, 23$, require u and opposite label $j+1$ before the pivot, and the first j opposite labels and first two chosen-side labels after it. The opposite gap then has length exactly j , the chosen gap has length at least two, and the loss is at least a_{j+1} . These events are disjoint for distinct j .

For a witness outside the 24 opposite labels, the probability is

$$p_j = \frac{2}{(j+5)(j+4)(j+3)}. \quad (3.6)$$

If u is opposite label $i + 1$, where $0 \leq i \leq 23$, it is

$$p_{ij} = \begin{cases} \frac{2}{(j+5)(j+4)(j+3)}, & j < i, \\ \frac{1}{(j+4)(j+3)}, & j = i, \\ 0, & j > i. \end{cases} \quad (3.7)$$

Indeed, for p specified distinct labels before the pivot and q specified distinct labels after it, the exact probability is

$$\frac{p! q!}{(p + q + 1)!}. \quad (3.8)$$

This is a relative-order count, with no independence assumption on the label events. Designating the outside row by $i = 24$, all 25 rows satisfy the exact certified inequality

$$\sum_{j=0}^{23} p_{ij} a_{j+1} \geq \delta. \quad (3.9)$$

4 The simultaneous-cap remainder

The two directional losses leave a further gain when both caps shrink. The identity

$$(1 + l + R)(1 + L + r) - KT = (L - l)(R - r) \geq 0 \quad (4.1)$$

gives the nonnegative remainder

$$\begin{aligned} J &:= \log \frac{(1 + l + R)(1 + L + r)}{KT} \\ &= \log \left(1 + \frac{(L - l)(R - r)}{KT} \right) \geq 0. \end{aligned} \quad (4.2)$$

Consequently,

$$\log X_m \leq \log K - D_L - D_R - J. \quad (4.3)$$

Lemma 3 (Simultaneous saving). *If $L, R \geq 2$ and $l, r \leq 1$, then $J \geq \log(16/15)$.*

Proof. We have $T \leq 3$, and $5(L - 1)(R - 1) \geq 1 + L + R = K$. Hence

$$\frac{(L - l)(R - r)}{KT} \geq \frac{(L - 1)(R - 1)}{3K} \geq \frac{1}{15}.$$

Apply the monotonicity of the logarithm in (4.2). The local bound is attained at $L = R = 2$ and $l = r = 1$. $\square$

For regular second-step witnesses in both directions, require the witnesses before the pivot and the four nearest side labels after it. Distinct witnesses avoiding these four labels give probability $2!4!/7! = 1/105$; coincident witnesses give $1!4!/6! = 1/30$. In either case the expected joint saving is at least ε , because

$$\frac{1}{105} \log \frac{16}{15} \geq \varepsilon. \quad (4.4)$$

A witness may instead coincide with one of the first two opposite labels. This prevents the simultaneous event but increases its own directional saving. The two relevant lower bounds are

$$\frac{a_1}{12} \geq \delta + \varepsilon, \quad \frac{a_1}{30} + \frac{a_2}{20} \geq \delta + \varepsilon. \quad (4.5)$$

The other direction retains its δ saving. Thus every regular cross-direction witness configuration gives combined saving at least $2\delta + \varepsilon$, including these collisions.

5 Exceptional vertices and charging

The joint bonus is an aggregate bound, since it cannot be assigned to each man separately. For each direction, define the noncycle set

$$B = \{m : g^2(m) \neq m\}.$$

Let D contain the exceptional men whose chosen side has at least two partners, with $g^2(m) = m$ and $g(b) \in \{m, g(m)\}$, where b owns the second partner.

Lemma 4 (Two-to-one charging). *For each direction, $|D| \leq 2|B|$.*

Proof. Charge an exceptional man m to its second-partner owner b . The neighbor-owner activity lemma gives $g(b) \neq b$, and the exceptional relations imply $g^2(b) \neq b$, so $b \in B$. Conversely, a man charging a fixed b must be either $g(b)$ or $g^2(b)$. Each fiber therefore has at most two elements. $\square$

Each directional mean loss is nonnegative, at least δ outside D , and at least $3(\delta + \varepsilon)$ on B . Short sides use (3.4); regular sides use the event rows of Section 3. A man regular on both sides has the combined gain from Section 4. If either side is short, its stronger saving already covers that combined target.

Put $S_m = \mathbb{E}_\pi[D_L + D_R + J]$. Write b_L, b_R and d_L, d_R for the indicators, at m , of the two noncycle and exceptional sets. The resulting pointwise accounting inequality is

$$S_m \geq 2\delta + \varepsilon + 2(\delta + \varepsilon)(b_L + b_R) - (\delta + \varepsilon)(d_L + d_R). \quad (5.1)$$

Summing over men and applying Lemma 4 separately in each direction cancels the possible deficits:

$$\sum_m S_m \geq n(2\delta + \varepsilon), \quad 2\delta + \varepsilon = \frac{23657}{500000} = 0.047314. \quad (5.2)$$

This estimate includes the cost of every exceptional vertex.

6 Exact constants and the final bound

Combining the gap estimate, (4.3), and (5.2) gives, for every stable reference μ ,

$$\mathbb{E}_\pi \left[\sum_m \log X_m(\mu, \pi) \right] \leq n(B_{512} - 2\delta - \varepsilon). \quad (6.1)$$

The remaining numerical comparison is exact:

$$B_{512} - 2\delta - \varepsilon < \log \frac{1589}{500}. \quad (6.2)$$

To certify the event bounds and this comparison, the development uses finite positive partial sums for the logarithm. For $0 \leq x < 1$ and an integer $t \geq 1$,

$$\log \frac{1+x}{1-x} = 2 \sum_{k=0}^{t-1} \frac{x^{2k+1}}{2k+1} + R_t(x), \quad 0 \leq R_t(x) \leq \frac{2x^{2t+1}}{(2t+1)(1-x^2)}. \quad (6.3)$$

For a_j , take $x = 1/(2j+3)$; six positive terms give the rational floors used in the 25 event rows. For the final comparison, take

$$x = \frac{76987}{3254987}, \quad \frac{1+x}{1-x} = \frac{1665987/500000}{1589/500}. \quad (6.4)$$

Four terms and the rational remainder bound certify that the logarithm of this ratio is less than $2\delta + \varepsilon$. Together with (2.4), this proves (6.2). All comparisons are proved in Lean by exact arithmetic.

Proof of Theorem 1. The empty stable family is immediate. Otherwise, average (6.1) over μ and use Lemma 1. Since $n > 0$, the strict inequality (6.2) and monotonicity of the exponential give

$$\begin{aligned} |\mathcal{S}(I)| &\leq \exp(n(B_{512} - 2\delta - \varepsilon)) \\ &< \exp\left(n \log \frac{1589}{500}\right) \\ &= \left(\frac{1589}{500}\right)^n. \end{aligned}$$

Taking the maximum over profiles completes the proof. □

A The Lean statement and verification

The exact public declaration in `JointEntropyUpperBound.lean` is:

```
theorem stableCount_lt_3178_pow {n : ℕ}
  (hn : 0 < n) (I : Profile n) :
  (stableCount I : ℝ) < (1589 / 500 : ℝ) ^ n
```

It is in namespace `SmpMax.General`. The count is coerced to $\mathbb{R}$; $1589/500$ is a rational expression in $\mathbb{R}$, not a floating-point approximation. The exact statement check and definition printout are in `lean/Check3178.lean`.

The final theorem’s axiom dependencies are exactly `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorry`, `admit`, new axiom, or `native_decide` in the packaged general proof sources. Numerical validation scripts and external SAT certificates are not assumptions of the theorem.

The bundle contains 56 unchanged general proof modules. The target’s transitive local dependency closure comprises 45 modules; the remaining modules retain earlier completed bounds. Lean is pinned to version 4.33.1, and the exact Mathlib and transitive dependency revisions are recorded in `lean/lake-manifest.json`.

From the bundle root, with Python 3.9 or newer and the Lean toolchain manager `elan` installed, run:

```
python3 tools/verify.py --fetch
python3 tools/verify.py --kernel all
```

The first command obtains the pinned dependencies, builds all 56 source modules, checks the exact declaration, audits 112 general statements, and independently replays the final module. The second also replays all 45 local target-dependency modules. Omit `--fetch` when dependencies are already present. For a file-identity check without Lean:

```
python3 tools/verify.py --sources-only
```

The retained verification was performed on 24 September 2026: all 56 modules rebuilt, all 112 audits passed, and all 45 modules replayed successfully. It reused the pinned dependency cache; the download step was not repeated. This 25 September edition changes the presentation only. The Lean sources, project configuration, verifier, and formal verification logs are byte-identical to that checked bundle. The logs and command results are in `evidence/verification.json` and its adjacent files.

B Guide to the formal development

All module names below are relative to `lean/SmpMax/General/` and have the suffix `.lean`. The complete source list and dependency order are in `SOURCE_MAP.md` and `source-index.json`.

Definitions and revelation. `Definitions`, `MatchingRevelation`, and `ExtendedRevelation` define the full stable family and establish the finite revelation inequality in Section 2.

Common-gap bound. `PartnerIntervals`, `PermutationCounting`, `LogTail`, and `RefinedLogBound` prove the finite gap estimate. `RefinedLogCertificate` checks its exact constant; `LongPaddedOwners` and `LongPaddedSupport` supply the longer padded-list interfaces.

Propagation and event rows. `SuccessorPropagation`, `BidirectionalPropagation`, `SupportCaps`, and `PartnerFrame` connect exclusions to actual compatible supports. `LongDirectionalSaving`, `LongEventRows`, and `LongerLogCertificate` establish the directional events and all 25 certified rows.

Joint savings and charging. `JointGapSaving`, `JointGapGeometry`, and `JointDirectionalSaving` cover the simultaneous remainder and witness collisions. `SuccessorCharging`, `JointCharging`, and `JointMatchingSaving` establish the global saving, including exceptional vertices.

Exact final comparison. `JointLogCertificate` proves (6.2); `JointEntropyUpperBound` assembles the support estimates and entropy inequality into Theorem 1.